

PURBANCHAL UNIVERSITY

2023

4yrs./3yrs. Bachelor of Computer Application (BCA)/Third Semester/*Final*
Time: 03:00 hrs. Full Marks: 60 /Pass Marks: 24

BCA203CO/BCA273CO: Data Structure and Algorithm (*New/Back*)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

- 1(a) Explain doubly linked list with its advantages. 6
- (b) Write an algorithm to insert a node as a first node in a linked list. 6
2. Discuss different types of tree. Explain different tree traversal algorithm with proper illustrations. 4+8
3. Define graph. Discuss different types of graph with example. Explain graph adjacency matrix representation with proper example. 2+6+4

Group B

Answer SIX questions.

6×6=36

4. What is the need of data structure? Discuss different operations of stack with example. 2+4
5. What are different types of queue? Explain advantage of circular queue over linear queue. 3+3
6. Define recursion and iteration with examples, and advantages and disadvantages.
7. What is spanning tree? Explain Minimal Spanning Tree with

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PURBANCHAL UNIVERSITY

2023

4 Years Bachelor of Computer Application (BCA)/Third Semester/Final
Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

BCA205CO/BCA275CO: User Interface Design (New/Back)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. What do you mean by goal-directed design? Explain the models of interface design, and discuss the most preferable one with appropriate reasons. 2+10
2. What are the menu design issues? Discuss the different types of dialog box used in user interface. 2+10
- 3(a) What are bounded and unbounded gizmos? Explain combo box. 3+3
- (b) Define branding. Write about the misconceptions of modern file system. 2+4

Group B

Answer SIX questions.

6×6=36

4. What is 'fear of interoperability'? Write the pros and cons of interoperable system.
5. Explain resizing, reshaping and repositioning with appropriate illustrations.
6. What is 'flow' in software behaviour? Explain task coherence.

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2023

4yrs./3yrs. Bachelor of Computer Application (BCA)/Third Semester/Final
Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

BCA202CO/BCA271CO: Computer Architecture (New/Back)

Candidates are required to give their answers in their own words as far as practicable.

All questions carry equal marks. The marks allotted for each sub-question is specified along its side.

Group A

Answer TWO questions.

2×12=24

- 1(a) What is register? Describe different types of registers for basic computer organization. 2+4
- (b) Describe the stored program organization for basic computer organization. 6
- 2(a) What are the methods of designing control unit? Describe about microprogram control unit with the help of diagram. 2+4
- (b) Write the comparison between Hard-wired control unit and Microprogram control unit. 6
- 3(a) What is multiprocessor? Describe the characteristics of multiprocessors. 2+4
- (b) Depict the differences between tightly and loosely coupled multiprocessor. 6

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(2)

- (b) Compare between Synchronous and Asynchronous data transfer. 4
- 8(a) Explain about memory hierarchy with diagram. 4
- (b) What is cache memory? Describe the operation of cache memory. 4
- 9(a) Describe the working mechanism of arithmetic pipelining with suitable examples and diagrams. 4
- (b) Differentiate between array processing and vector processing. 4
- 10. Explain four kinds of interconnection network with examples.
- 11. Write short notes on any TWO: 2×4=8
 - (a) Instruction cycle
 - (b) Cache coherence
 - (c) Memory mapping

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4yrs/3yrs Bachelor of Computer Application (BCA)/Third/Semester/Final

Time: 03:00 hrs.

Full Marks: 80 / Pass Marks: 32

BCA204CO/BCA270CO: System Analysis and Design (New/Back)

BCA204CS: System Analysis and Design (Old)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. What do you mean by system analysis and design? Explain its significance in software development? Describe the system development life cycle in brief.
2. Define process and modeling. Explain DFD and draw DFD for Library Information System. (Make necessary assumptions).
3. What is logic modeling? Explain, with a suitable example, how decision table help during logic modeling.

Group B

Answer SEVEN questions.

7×8=56

4. Why system design is considered more challenging than other phases of system development? Explain how system design

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(2)

10. What are the advantages of modular design? Explain the importance of coupling and cohesion in effective modular design.
11. What are structured charts? Explain its use in system development process.
12. Write short notes on any TWO: 2×4=8
(a) Entity Relationship Diagram
(b) Physical database design
(c) Use-case modeling

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(4)

Makalu Inc. expects sales of 600,000 units for the year, purchased by the firm for Rs. 10 per unit. The order cost is Rs. 30 and Makalu's carrying cost percentage of the inventory value has been estimated @ 20%.

Required:

- What is the economic order quantity?
- How many orders Makalu should place each year?
- What is the total cost of inventory at EOQ?

Write short notes on any TWO:

- Cash book
- Cash flow statement
- Rules of double entry system
- Accounting information technology

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Time: 03:00 hrs. Full Marks: 60 /Pass Marks: 24

BCA201MS: Financial Accounting (New Course)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

- What is financial management? Describe the objectives of business firms. Wealth maximization is superior than profit maximization? Justify.
- The following ledger balances of a company is given as on 31.12.2021.

Debit	Rs.	Credit	Rs.
Plant and Machinery	1,25,000	Purchase return	4,500
Drawings	40,000	Creditors	40,000
Land & Building	80,000	Provision for doubtful debts	5,000
Salaries	32,450	Rent received	8,000
Cash in hand	21,250	Sales	2,75,000
Motor vehicles	53,300	Bank overdraft	15,500
Investment	48,000	Capital	4,00,000
Goodwill	38,000	Commission	4,000
Sundry debtors	60,400	Bills payable	36,000
Advertising	20,500		
Purchases	1,40,000		
Legal charges	12,500		
Carriage inward	9,600		
Wages	23,000		
Rent	30,000		
Stock on 1.1.2002	42,000		
Sales commission	12,000		
Total	788,000		788,000

You are required to prepare Trading Account, Profit and Loss Account and Balance Sheet after taking into following adjustments:

Contd....

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4. Mr. Messy wishes to determine how much money he will have at the end of five years, if he deposits Rs. 1,000 annually in a saving account paying 8% annual interest. The deposits will be made at the end of each of the next five years and show the time line.
5. From the following data, calculate:
- Gross profit ratio
 - Current ratio
 - Inventory turnover ratio
 - Liquid ratio
 - Assets turnover ratio
 - Net profit ratio

Contd....

- (ii) Advance tax of Rs. 7,000 is payable in February and May.
 - (iii) Period of credit allowed to customers is one month and suppliers two months.
 - (iv) Lag in payment of office expenses and wages is one month.
7. What do you understand by dividend policy? Discuss the factors affecting dividend policy.
8. Explain the concept of optimal capital structure with suitable example.
9. How is cash from operating activities calculated under direct method? Illustrate.

Contd...

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